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REVIEW

Efficacy of laser acupuncture for patients with knee osteoarthritis: a systematic review and meta-analysis of randomized controlled trials

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Abstract

Laser acupuncture (LA) has been increasingly investigated as a non-invasive therapy for knee osteoarthritis (KOA), yet its clinical efficacy remains uncertain. To evaluate the effectiveness and safety of LA for pain, function, and mobility in KOA through a systematic review and meta-analysis. Randomized controlled trials (RCTs) comparing LA with sham, placebo, electroacupuncture, or standard care were identified from major databases. Outcomes included pain (VAS, WOMAC-Pain, NPRS, PI), function (WOMAC-Function), and range of motion (flexion). Risk of bias was assessed using RoB 2, and heterogeneity explored through subgroup analyses. Thirteen RCTs of 611 participants were included. LA did not significantly improve WOMAC pain (SMD 0.04; 95% CI −0.36 to 0.45), pVAS (SMD 0.12; 95% CI −0.91 to 1.15), or NPRS (SMD −0.23; 95% CI −0.58 to 0.12). Function showed a nonsignificant trend toward improvement (SMD −0.32; 95% CI −0.94 to 0.30), and knee flexion outcomes were also nonsignificant (SMD 0.10; 95% CI −0.39 to 0.59). Substantial heterogeneity ($I^2 > 70\%$) was observed across studies, likely related to differences in wavelength (650–904 nm), dosage, and acupoint protocols. Funnel plots did not suggest marked publication bias. Current evidence does not demonstrate statistically significant benefits of LA over comparators for pain, function, or flexion in KOA. Considerable heterogeneity and variability in laser parameters limit firm conclusions. Large, standardized RCTs with optimized parameters are needed to confirm efficacy and guide clinical application.

Clinical trial number: Not applicable.

Keywords Knee osteoarthritis · Laser acupuncture · Pain management · Photobiomodulation · Systematic review · Meta-analysis

Introduction

Osteoarthritis (OA) is a worldwide, highly prevalent chronic joint disease that causes pain, disability, and loss of function. From 1990 to 2019, there was a substantial 114.5% rise in the global incidence of years lived with disability attributed to OA [1, 2]. Prevalence cases and the Age-Standardized Rates of OA has shown a consistent global and regional increase over time. The occurrence of OA becomes more prominent with advancing age and demonstrates a higher prevalence among females, along with variations in geographic distribution and affected anatomical sites. Among the joints affected by OA, the knee stands as the most commonly impacted, followed by the hand [3–6]. There is no non-surgical intervention that can prevent, halt, or delay OA progression. The population of the elderly has comorbidities that may lead to increased side effects due to drug interactions [7]. Moreover, available medications, such as nonsteroidal anti-inflammatory drugs, have been associated with an increase in the risk of myocardial infarction, drug dependency with increasing the dosage, liver and kidney metabolic damage, or death from cardiovascular causes [8–10].

Acupuncture is renowned for its minimal side effects and its widespread application in the management of chronic pain [11]. Acupuncture has garnered increasing attention as an alternative strategy for knee osteoarthritis (KOA), supported

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by several studies demonstrating its efficacy [12, 13]. Moreover, a systematic review by the Osteoarthritis Research International Society (OARSI) stated that acupuncture may have a symptomatic benefit in patients with KOA [6, 14]. Acupuncture involves the use of fine needles and knowledge of human anatomy, physiology, and pathology, plus the principles of evidence-based medicine. Acupuncture, though highly used, has the possibility of side effects of pain at acupuncture points, infection, and syncope [15].

Laser acupuncture (LA) stands out as one of the most popular techniques amidst the various approaches available within the field of acupuncture. LA is non-invasive and has been used in China for about 20 years to treat KOA [12, 16]. When compared to traditional needling, it is better suited for stimulating tough points such as auricular acupuncture points [17–20]. Although LA uses the same photobiomodulation (PBM) as Low-Level Laser Therapy (LLLT), the two modalities differ in application as LLLT targets local pathological tissues, while LA delivers laser simulation at acupoints to modulate neurophysiological pathways [21–24]. Neuroimaging and electrophysiological studies further suggest that low-intensity acupuncture may influence central nervous activity, including the activation of the reticular activating system [25].

This study aimed to provide an updated perspective on LA by performing a systematic review and meta-analysis, building upon the findings of the previous systematic reviews and meta-analyses [26–29]. We assessed the efficacy of LA in alleviating symptoms and increasing overall function in patients with KOA. This meta-analysis incorporates more recent RCTs and focuses specifically on LA applied at acupuncture points, providing a clearer and more up-to-date evaluation of its clinical effects in KOA.

Methods

Protocol

This meta-analysis was reported according to the Preferred Items for Systematic Review and Meta-Analysis (PRISMA) statement and is registered with PROSPERO (CRD42023448524) [30].

Searching strategies

Library, MEDLINE, CINAHL, and PEDro, encompassing the period from the inception of each database until August 2023. The search utilized a range of keywords such as “acupuncture”, “osteoarthritis”, “knee osteoarthritis”, “laser acupuncture”, and “randomized controlled trials” (supplementary files). Two reviewers, VnK and GL thoroughly assessed the titles and abstracts, followed by a detailed examination of the full texts of potentially eligible RCTs to determine their suitability for inclusion in the meta-analysis.

The search strategy employed for the database is presented in Fig. 1.

Study selection

Inclusion and exclusion criteria

Study: This study exclusively focused on RCTs investigating the efficacy of LA for KOA. Only trials meeting the following criteria were considered: a sample size exceeding 20 patients, participants aged 40 and above, a confirmed diagnosis of KOA, and LA after knee arthroplasty was added. This research focused on using RCTs to investigate the efficacy of laser acupuncture for osteoarthritis in the knee. Observational studies, animal studies, reviews, and protocols were excluded from the analysis [30, 31].

Participants: Patients who were diagnosed with KOA based on radiographic evaluation or in accordance with the clinical criteria outlined in the American College of Rheumatology guidelines [19, 31].

Intervention: The experimental group underwent LA treatment, whereas the comparator received sham LA, placebo treatment. Trials that included LLLT at acupoints were included. Excluded from the analysis were trials that specifically contrasted true acupuncture with sham acupuncture.

Outcome measures

The primary emphasis involved the analysis of key results, centered around the examination of pain indicators. These indicators were evaluated using the Western Ontario and McMaster Universities Arthritis Index (WOMAC) [32, 33] the pain-on-motion visual analog scale(pVAS) and Numerical Pain Rating Scale (NPRS). Secondary outcomes encompassed WOMAC function and flexion.

Literature selection

Based on the search approach, the articles were sought and evaluated based on the specified criteria for inclusion and exclusion. Replicates were eliminated, and in order to make the final selection, the complete texts were extensively examined and analyzed by two authors (VnK, GL).

Data analysis

The Review Manager software, version 5.4.1 [34], was utilized to conduct a meta-analysis. In this analysis, as all

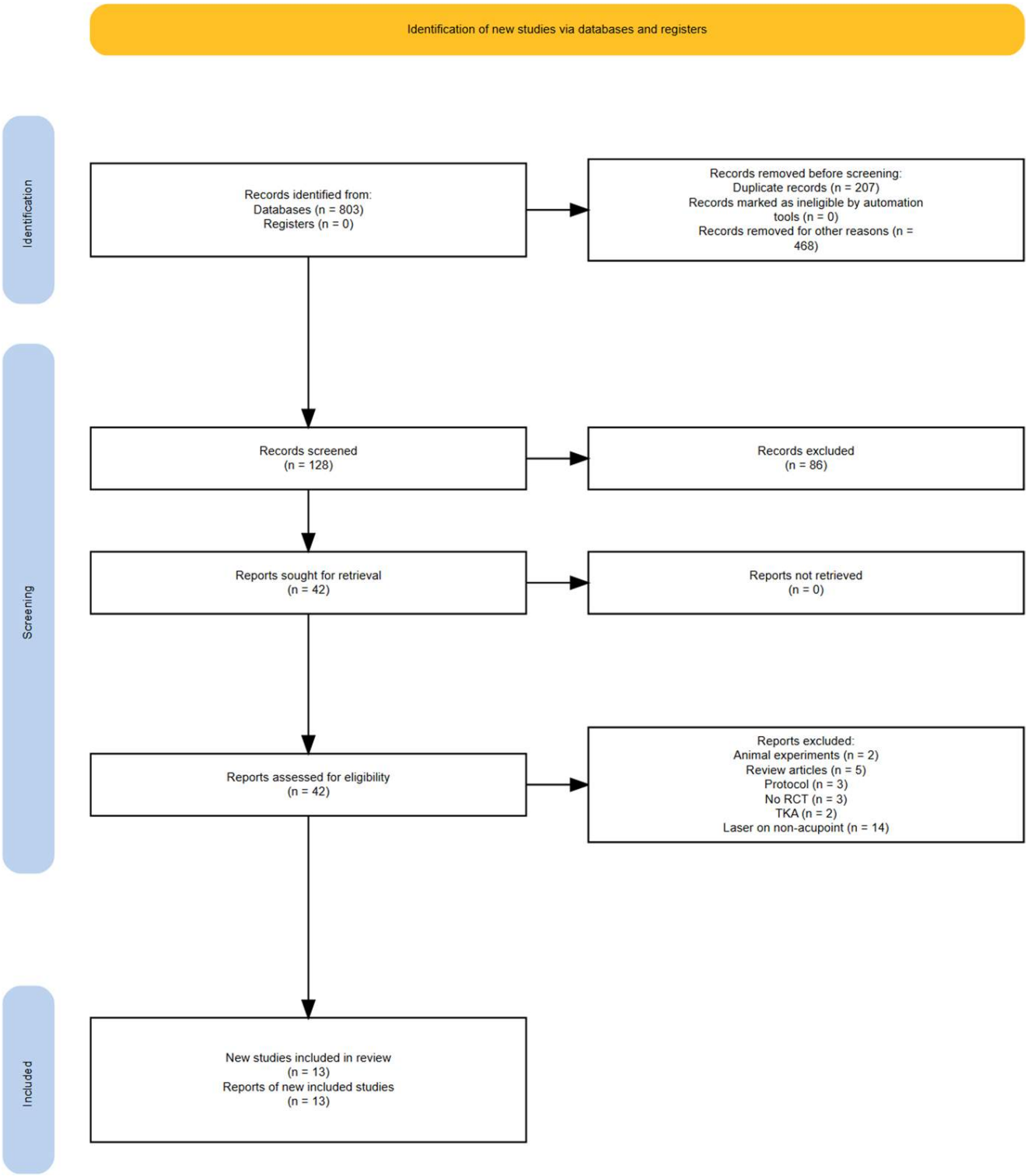


Fig. 1 PRISMA 2020 flow diagram for systematic reviews

the results encompassed continuous data, we computed the standard mean difference (SMD) along with a 95% confidence interval (CI), utilizing the means and standard deviations. An SMD in the negative range indicated a preference for LA over the control intervention, and the opposite was true. To evaluate clinical relevance and statistical heterogeneity, the chi-square test ($p < 0.05$) and I^2 statistics ($\geq 50\%$).

Heterogeneity was interpreted primarily using I^2 statistic, chi-square results were reported descriptively. were employed, respectively. For the primary analyses, a random-effects model was adopted. Missing data were handled by extracting available post-intervention means and standard deviations. If required data were not reported and could not be obtained from study authors, they were excluded from the specific pooled analysis. Publication bias was assessed through funnel plots. Egger's test was not performed because fewer than ten studies were available per outcome, consistent with Cochrane Handbook guidance that asymmetry tests should not be used with small study numbers due to low statistical power [Cochrane Handbook, Sect. 13.3.4.4].

Assessment of risk of bias

The Cochrane Risk of Bias 2 (RoB 2) tool for randomized trials was employed to evaluate the potential bias in each trial included, covering five domains: (1) bias arising from the randomization process, (2) bias due to deviations from intended interventions, (3) bias due to missing outcome data, (4) bias in measurement of the outcome, and (5) bias in selection of the reported result. Judgments ("low risk," "some concerns," or "high risk") were assigned at the outcome level, as recommended by the Cochrane Handbook. The RoB was evaluated independently by two reviewers, VnK and GL, and a consensus was established before the final assessment was determined [35–37].

Summary measures

Effect sizes were assessed using standardized mean differences, denoted by Hedges' g (unbiased). This correction accounts for the tendency of studies with limited sample sizes to overestimate the true population effect. To facilitate comparison, we also presented the mean difference (MD), as MD = M1 − M2. Various formulations of the standardized mean difference (SMD) exist. In review manager, the employed formulation is Hedges' adjusted g, closely resembling Cohen's d, but with an adjustment for small sample bias [38, 39].

Hedges' formula

$$g = \frac{M1 - M2}{SDPooled}\left(1 - \frac{3}{4N - 9}\right)$$

Where:

- M1: mean response for the patient group,
- M2: mean response for the control group, and.
- N: total sample size, both patient and control groups [39].

The pooled standard deviation (SD pooled) is calculated using the formula:

$$SDPooled = \sqrt{\frac{(N1-1)SD1^2+(N2-1)SD2^2}{(N1+N2)-2}}$$

where:

- N1: sample size of the patient group,
- N2: sample size of the control group,
- SD1: standard deviation of the mean for the patient group, and.
- SD2: standard deviation of the mean for the control group.

Results

Study selection

The initial database search yielded 803 records. After duplicate and irrelevant articles were removed, 47 full-text articles underwent evaluation for eligibility, resulting in the exclusion of 34 studies that didn't align with our inclusion criteria upon thorough review (supplementary materials). Ultimately, 13 studies met the eligibility criteria and were included in the analysis (Fig. 1).

Characteristics of the included trials

The Meta encompassed a total of thirteen trials. Figure 1 illustrates the screening and selection process employed for these studies. Table 1 provides comprehensive information about the participants' characteristics, acupuncture details, control groups, treatment regimens, outcome measures, and results.

The trials exhibited diversity in the duration and frequency of LA sessions, spanning from once a week to three times a week over a duration of 2 to 13 weeks. RoB judgements were applied at the outcome level using the Cochrane RoB 2 tool. The collective RoB across the trials highlights the main bias origins associated with participant and personnel blinding. Figure 2 further illustrates the distinct risk of bias for each trial, with seven displaying low risk, five demonstrating some concerns, and one displaying high risk.

Results of meta-analysis

Primary outcomes

WOMAC pain scores from eight RCTs showed the pooled effect showed no significant difference between Treatment

Table 1 Characteristics of studies. Characteristics of trials included in the meta-analysis. pVAS: pain on motion with Visual Analogue Scale, NRS: Numerical Rating Scale of pain, TUGT: Timed Up and Go Test, MAT: Magnetic Auriculotherapy, LAT: Laser Auriculotherapy, KC: Knee Circumference, NHP: Nottingham Health Profile total score, PI: Post Intervention, VAS: Visual Analogue Scale, SKFS score: Saudi Knee Function Scale, MPQ: McGill Pain Questionnaire, WAI: Working Alliance Inventory, MHLC: Multidimensional Health Locus of Control, NPRS: Numeric Pain Rating Scale

No.	Study	Sample size(F: M)	Mean age	Intervention/Acupoint	Laser parameters	Treatment sessions	Outcomes	Results
1	Liao et al. 2020	30(27:3)	Active laser 70.53 ± 6.89, placebo laser 69.73 ± 6.91	Laser acupuncture (15), placebo laser (15) at acupoints SP9, SP10, EX-LE2	Dual-frequency LLLT (780 nm, 50 mW + 830 nm, 30 mW; 216 J total, 15 min continuous	3 times a week for 4 weeks	pVAS, PPT, Lequesne index	Significant reduction in pain (VAS), improved PPT & Lequesne index in ALT vs. PLT
2	Lafta et al. 2018	69(21:48)	56.65 ± 8.56 57.36 ± 6.85 51.39 ± 8.57	Laser Acupuncture Group (23) at acupoints SP10, ST34, ST35, GB34, Xiyan, SP9	10 min, 18 J, 830 nm 45 J/15 cm2	10 sessions, 5 times a week for 10 min 10 sessions, 2 weeks, Superficial heat therapy and isometric knee exercises for 15 min every,10 sessions in 2 weeks PI 2 weeks	WOMAC, McGill Pain Questionnaire, NPRS, Flexion, Extension	Laser acupuncture and conventional laser are more effective in reducing pain and increasing functional activity in patients with KOA
3	Mohammed et al.2018	40(6:34)	Active laser 55.25, control 50.11	Active laser (20) sham (20) at acupoint ST36, SP10, GB, ashi	60 s, 5.4 J, 808 nm	4 weeks, 12 sessions	Pvas, Beta-endorphin Substance P	Pain (VAS) was improved in all groups; however, the decrease in pain perception was greater in patients treated with laser acupuncture
4	Rees et al. 2017	40(20:20)	Actie laser 61.3 ± 11.36; placebo 63.0 ± 10.11	Active laser (20), placebo (20) at 13 acupoints	26 min, 18 J, 74.4Jcm2	12 sessions, 3 weeks follow-up after 8 and 12 weeks	WOMAC pain score; MPQ WOMAC function score WOMAC stiffness score WAI; MHLC	Laser acupuncture reduced KOA pain and stiffness and improve physical function
5	Helianthi et al. 2016	59(17:42)	69 ± 5	Active laser acupuncture $n = 30$ at point ST35, ST36, SP9, GB34 and EX-LE-4 Placebo laser $n = 29$ at ST35, ST36, SP9, GB34 and EX-LE-4	80 s (4 J)785 nm	10 sessions administered twice a week. Follow up 2 weeks	pVAS, Lequesne index	Active laser acupuncture group demonstrated a significant improvement in VAS scores and Lequesne as compared to the placebo
6	Suen et al. 20,162	43(2:41)	72.66 ± 7.04	Acupoint TF4, AH4, CO13, CO10, CO12, AT4. Group 1 Placebo LAT & MAT. Group2, LAT & Placebo MAT. Group 3 LAT + MAT. Group 4, Placebo	60 s 0.54 J/cm2 650 nm	6 weeks intervention	NRS, TUGT, and standard goniometer measurements during knee flexion and extension	The combination of MAT and LAT demonstrated a more potent treatment effect compared to the individual treatments of MAT or LAT and placebo group.
7	Hinman at el. 2014	282(143:139)	-	Control (71), Needle (70), Laser (71), Sham Laser (70) acupoints selected depending on clinical examination	20 min. 0.2 J	8 sessions, 2 weeks. follow-up At 12 weeks and 1 year	NPRS; WOMAC pain score, WOMAC function score, AQoL-6D; SF12	Needle and laser acupuncture had no long-term efficacy compared to sham laser acupuncture, despite providing short-term pain relief. sustained at 1 year

Table 1 (continued)

No.	Study	Sample size(F: M)	Mean age	Intervention/Acupoint	Laser parameters	Treatment sessions	Outcomes	Results
8	Al Rashoud et al. 2014	49(18:31)	Active laser 52±9; placebo 56±11	Active laser acupuncture (26), placebo (23) at five acupoints on affected knee (SP9, SP10, ST36)	40 s, 4 J/cm2 830 nm,	9 sessions, follow-up after the fifth and last session; Follow-up: 6 weeks, 6 months	Pain—VAS Function—SKFS score	Application of LLLT to specific acupuncture points plus exercise is effective in reducing pain
9	Alghadir et al., 2014	40(18:22)	Active laser 55.2±8.14, placebo laser 57±7.77	20 active, 20 placebo at 8 sites: medial, lateral, popliteal tendon area	GaAs diode laser (850 nm, 100 mW, 1 mm beam); 6 J/point, 8 points (48 J total)	2 times a week for 4 weeks	pVAS, WOMAC pain, stiffness, function	Improvement in pain, WOMAC pain & function, and ambulation in active vs. placebo
10	Zhao et al. 2010	40(35:5)	Active laser: 60.10±6.83, Sham laser: 59.40±6.15	active, 20 sham 31 at ST35	Combined 650 nm (36 mW)+10.6 µm CO_2 laser (200 mW, pulsed 40 Hz, 20 min; ~120 J CO_2 + 43.2 J 650 nm	3 times a week for 4 weeks	WOMAC pain, stiffness, function	Significant short-term pain reduction at 2 weeks; no long-term effect at 4 weeks
11	Shen et al. 2009	40(2:36)	Active laser: 60.10±6.83; placebo: 56.40±7.41	Active laser at acupoint ST35 $n=20$ Placebo laser at acupoint ST35, $n=20$	20 min, 650 nm	12 sessions, 3 times a week for 4 weeks.	WOMAC pain index score. WOMAC function score WOMAC stiffness score	Combined laser acupuncture with moxibustion improved the symptoms of OA
12	Yurtkuran et Al. 2007	55(2:53)	Active laser: 51.83±6.83; placebo: 53.48±7.13	Active laser acupuncture at acupoint SP9, $n=27$ Placebo acupoint SP9, $n=25$	120 s, 0.48j 904 nm	2 weeks, 10 sessions, 20 min each	WOMAC pain, WOMAC function score, WOMAC stiffness score KC; MTS; NHP	Compared to placebo, laser acupuncture demonstrated effectiveness in reducing periarticular swelling
13	Tascioglu et al. 2004	60(43:17)	Active laser: 62.86±7.32,59.92±7.59; placebo: 64.27±10	Active laser acupuncture at acupoint SP9, $n=20$ Placebo acupoint SP9, $n=20$ at 5 acupoints	GaAlAs diode laser (830 nm, 50 mW, continuous); 3 J/point (Group 1), 1.5 j(group2)	5 times a week for 2 weeks	WOMAC pain, stiffness, function; pVAS rest & movement	No significant differences among groups at week 3 or month 6



Fig. 2 Risk of bias analysis of included studies (**a**) Risk of bias graph; (**b**) Risk of bias summary

and Comparator, [n=418; SMD 0.04 95% C. I (−0.36,0.45); p=0.83; I^2=74%] Fig. 3 (a). The follow-up finding assessment from three studies indicated no significance for LA regarding its long-term effectiveness in pain relief, [n=302; SMD −0.11 95% C. I (−0.68, 0.47); p=0.71; I^2=73%] Fig. 5 (a).

The meta-analysis of pVAS scores from eight RCTs showed no significance in the pooled effect [n=371; SMD 0.12; 95% C. I (−0.91, 1.15); p=0.82; I^2=95%], Fig. 3(b). Additionally, the findings from the follow-up evaluation of four studies showed no significance in alleviating pain n=201; SMD 0.47 95% C. I (−1.19, 2.13); p=0.58; I^2=96%] Fig. 5(b).

NPRS data extracted from three RCTs indicated there was no significance in the pooled effect both after intervention and in the long-term follow-up assessment [n=188; SMD −0.23, 95% C. I (−0.58, 0.12); p=0.20; I^2=21%], as represented in Fig. 3(c). [n=128; SMD 0.00, 95% C. I (−0.35, 0.35); p=1.00; I^2=0%], Fig. 5 (c).

Secondary outcomes

WOMAC function data extracted from nine RCTs indicated that there was no significance in the pooled effect both after intervention and in the long-term follow-up assessment [n=464; SMD −0.32, 95% C. I (−0.94, 0.30); p=0.31; I^2=90%], [n=202; SMD −0.18 95% C. I (−1.06, 0.70); p=0.69; I^2=88%] Fig. 4 (a), Fig. 5 (d).

Two RCTs reported knee flexion ROM in KOA patients showed no significance in both analyses [n=65; SMD 0.10, 95% C. I (−0.39, 0.59); p=0.69; I^2=0%], [n=19; SMD −0.29, 95% C. I (−1.19, 0.62); p=0.54] Fig. 4 (b), Fig. 5 (e).

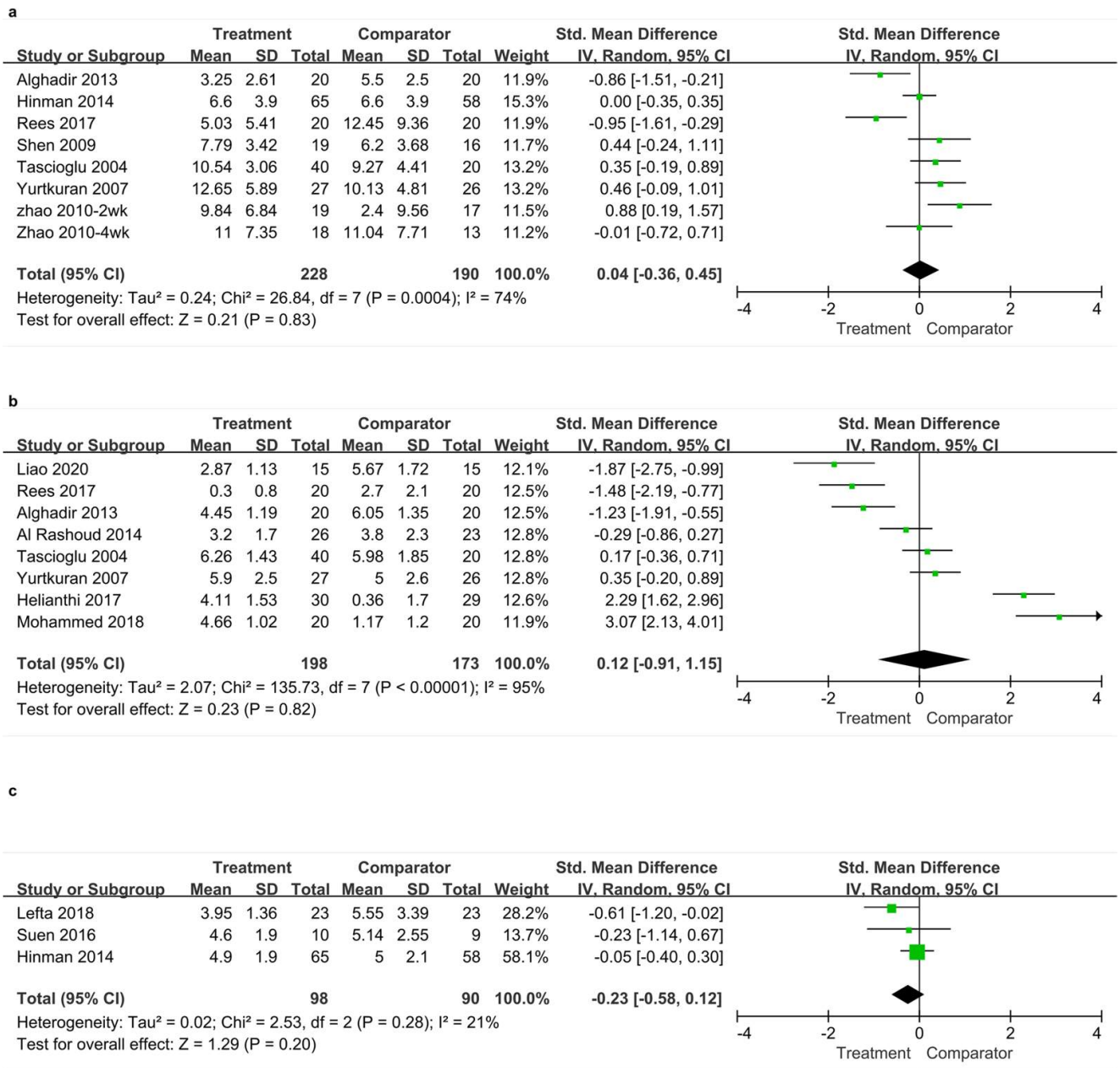


Fig. 3 Primary outcomes of the forest plot of laser acupuncture on (**a**) WOMAC pain, (**b**) pVAS, (**c**) NPRS

Follow-up

The pooled follow-up analysis across all outcomes included a total of 752 observations. The overall pooled effect showed no statistically significant difference between groups [$n = 752$; SMD 0.02; 95% CI (−0.44, 0.48); $p = 0.93$; $I^2 = 89\%$], with no evidence of subgroup differences ($I^2 = 0\%$), Fig. 5.

Funnel plots for the outcomes pain, WOMAC function, flexion, NPRS, and PI did not reveal substantial asymmetry, suggesting no major evidence of publication bias (Supplementary Figs. 1).

Discussion

This study provides comprehensive evidence from 13 randomized trials assessing the role of LA in KOA. Across studies, functional outcomes showed small, non-significant trends toward improvement, while pain results were inconsistent and did not reach statistical significance in pooled analyses. Although several individual trials reported within-group changes, these effects were not consistently replicated in the meta-analysis. Although analgesic effects remain more uncertain and heterogeneous across protocols [18, 23, 40–50].

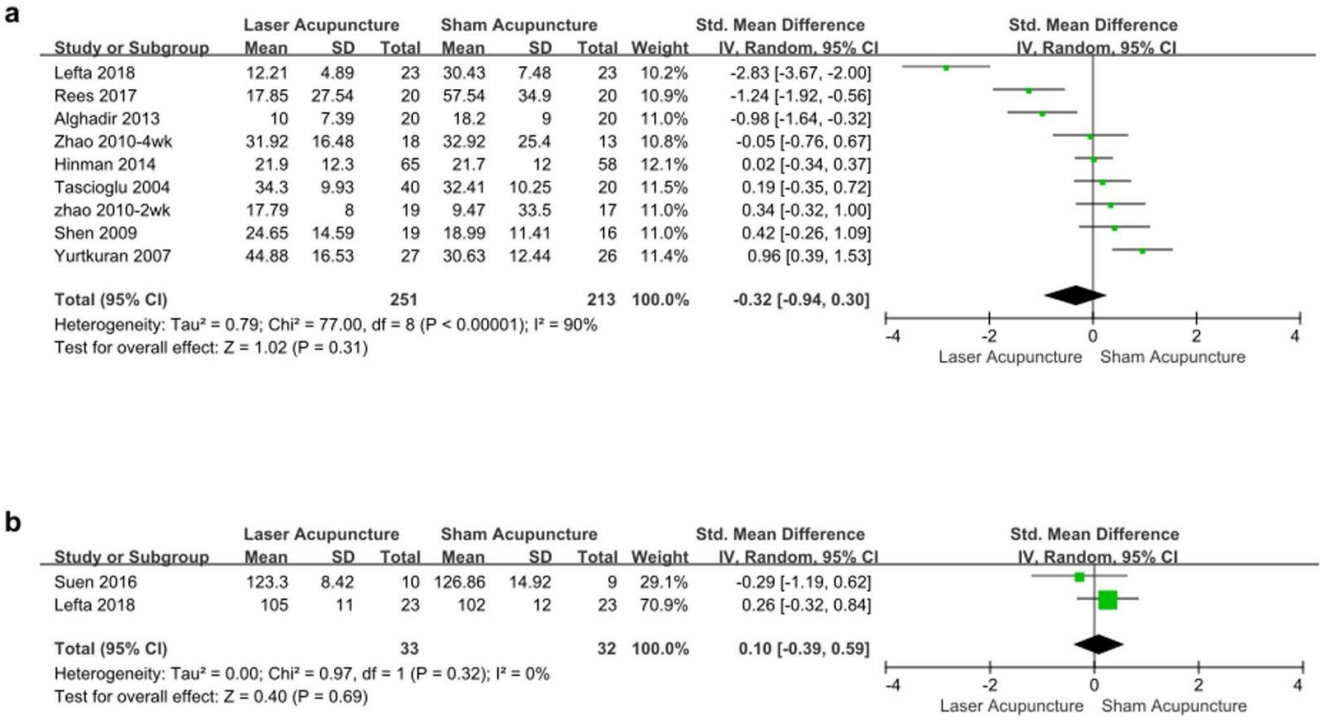


Fig. 4 Secondary outcomes for forest plot of laser acupuncture on (**a**) WOMAC function, (**b**) Flexion

A significant problem is the considerable heterogeneity among trials, since laser parameters exhibited extensive variation, including wavelengths ranging from 650 to 904 nm, along with discrepancies in power, irradiance, spot size, pulse mode, and session count. Previous PBM studies frequently report benefits with ~ 800–850 nm wavelengths, although our review did not perform dose–response or wavelength-specific analyses. Evidence from recent meta-analyses suggests that wavelengths above 1000 nm and power exceeding 100 mW are more likely to yield therapeutic benefit [51, 52]. Han et al. (2024) further demonstrated stronger outcomes with 810 and 785 nm devices in a pooled cohort of over 2000 patients [53]. Variability in acupoint protocols also contributed to heterogeneity, commonly used acupoints included GB34, ST36, ST34, and SP9, though protocols varied considerably [27, 54]. Comparator arms ranged from sham laser to placebo laser, which may have amplified expectancy effects and placebo responses. Placebo effects boost acupuncture's pain relief by activating endogenous opioids. Short-term, patients preferred the comparator over LA due to these beneficial placebo effects driven by positive expectations [55–58]. However, the long-lasting impact was not observed.

Despite this variation, our findings are broadly consistent with acupuncture and PBM literature. As shown by Stausholm et al. [29], using the World Association for Laser Therapy treatment guidelines to determine the laser dose per treatment point along the knee joint line has been associated with a noticeably positive outcome [30]. Some recent works have confirmed wavelength-dependent effects [52, 53, 59], while Allam et al. [60] reported functional and ROM improvements in a recent RCT. However, older reviews such as Huang et al. [61] were less conclusive, likely reflecting the small sample sizes and inconsistent laser protocols of earlier trials. Collectively, the present synthesis supports the notion that efficacy is not universal, but parameter-dependent, aligning with the broader PBM evidence base.

In clinical practice, LA can be considered a safe, non-invasive adjunct for KOA, particularly for patients with functional impairment or reduced ROM. Its benefits, however, appear highly parameter-dependent, emphasizing the need for adherence to consensus recommendations such as those of the World Association for Laser Therapy [52].

Limitations and future directions

This review has several limitations. First, substantial heterogeneity was present across trials, largely due to variations in wavelength, power, treatment duration, acupoint selection, and comparator design. Second, many included studies had small sample sizes and unclear or high risk of bias in one or more domains. Finally, publication bias could not be formally tested using Egger's method due to

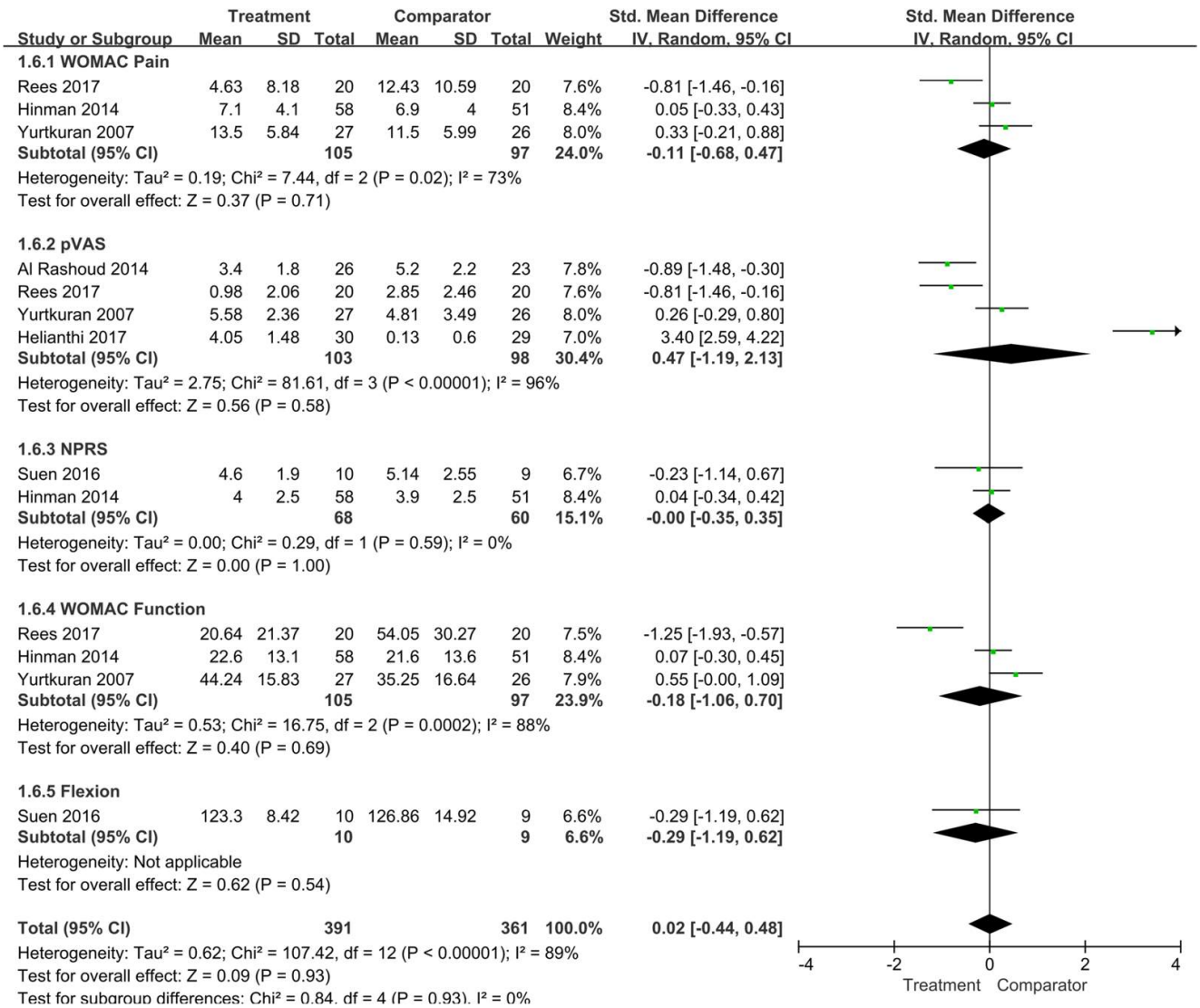


Fig. 5 Forest plot of the follow-up assessment

fewer than ten studies per outcome, which restricts certainty in interpreting asymmetry.

Future research should prioritize large, multicenter RCTs with standardized protocols for wavelength, power, energy density, pulse parameters, and treatment schedules. Comparative studies with EA, manual acupuncture, and sham controls are needed to clarify relative effectiveness. Extended follow-up will help establish durability, while incorporation of mechanistic biomarkers such as inflammatory mediators, laser acupuncture mechanism, and cartilage imaging would strengthen causal inference. Ultimately, well-designed trials bridging preclinical mechanistic insights with clinical outcomes are essential for defining the optimal role of LA in KOA management.

Conclusion

LA demonstrates potential for modest improvements in function and knee mobility in patients with KOA, although analgesic effects remain uncertain due to heterogeneous protocols and largely non-significant pooled results. Considerable variability in wavelength, power, treatment duration, and acupoint selection contributed to high between-study heterogeneity, underscoring the need for standardized intervention parameters. While LA appears safe and may serve as an adjunct for patients seeking non-pharmacological options, more rigorously designed, adequately powered randomized trials are required to confirm its clinical effectiveness and define optimal treatment protocols.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10103-026-04808-5.

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Data availability The data generated for the present study are available from the corresponding author upon reasonable request.

Declarations

Ethical approval PROSPERO (CRD42023448524).

Competing interests The authors declare no competing interests.

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